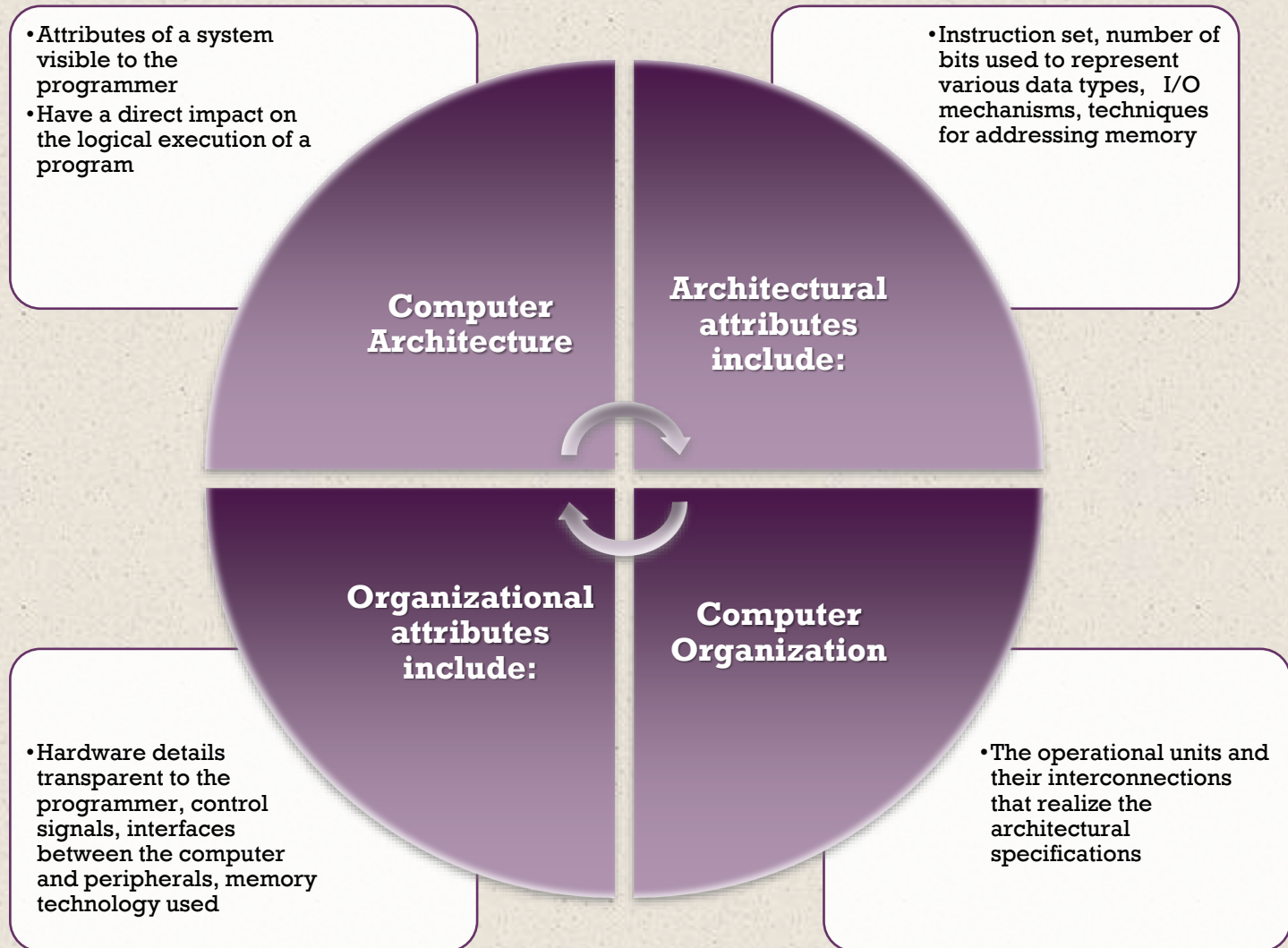


Chapter 1

+

Basic Concepts and Computer Evolution

Computer Organization



+ Structure and Function

- Hierarchical system
 - Set of interrelated subsystems
- Hierarchical nature of complex systems is essential to both their design and their description
- Designer need only deal with a particular level of the system at a time
 - Concerned with structure and function at each level
- Structure
 - The way in which components relate to each other
- Function
 - The operation of individual components as part of the structure



- There are four basic functions that a computer can perform:
 - Data processing
 - Data may take a wide variety of forms and the range of processing requirements is broad
 - Data storage
 - Short-term
 - Long-term
 - Data movement
 - Input-output (I/O) - when data are received from or delivered to a device (peripheral) that is directly connected to the computer
 - Data communications – when data are moved over longer distances, to or from a remote device
 - Control
 - A control unit manages the computer's resources and orchestrates the performance of its functional parts in response to instructions

Structure

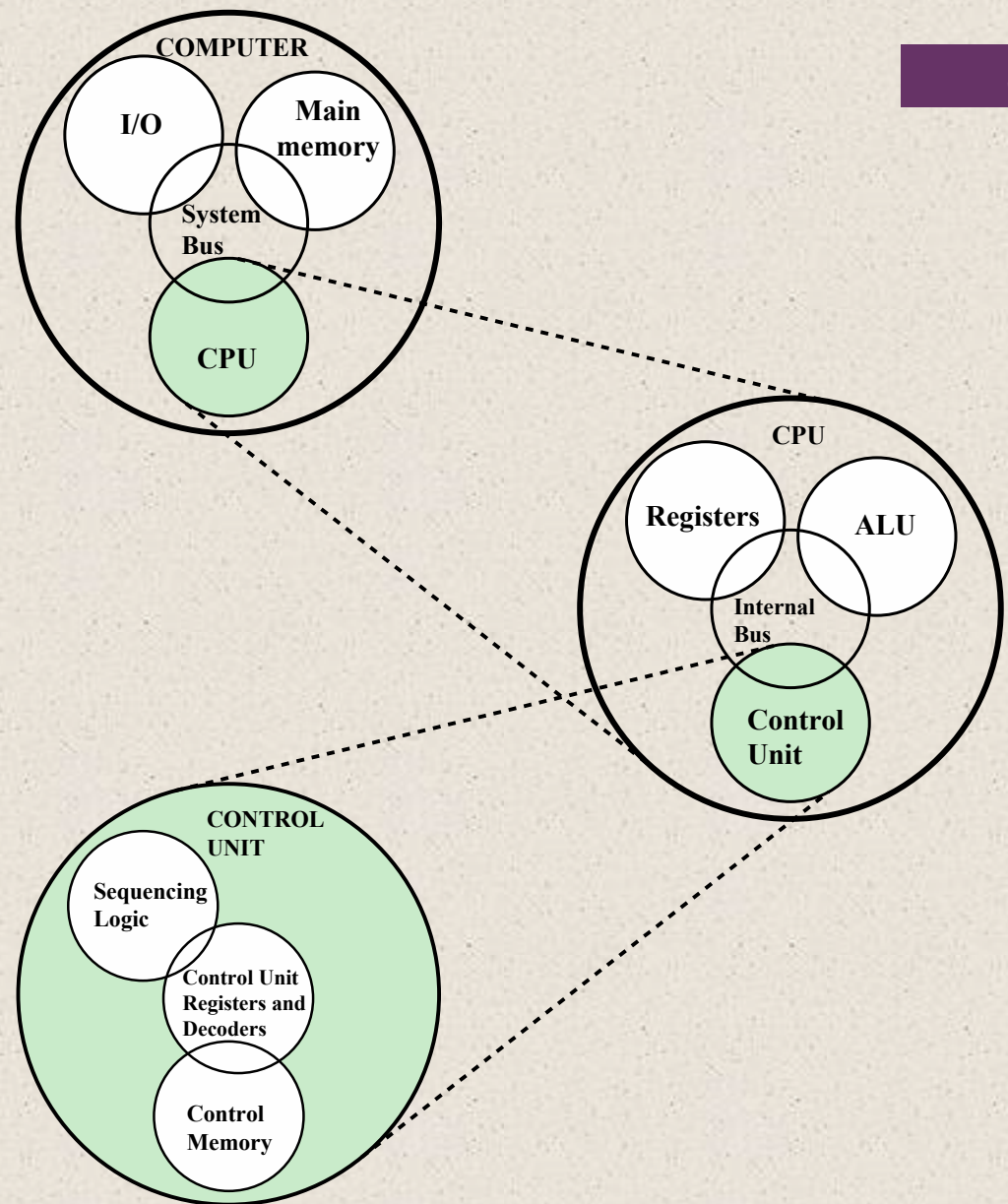
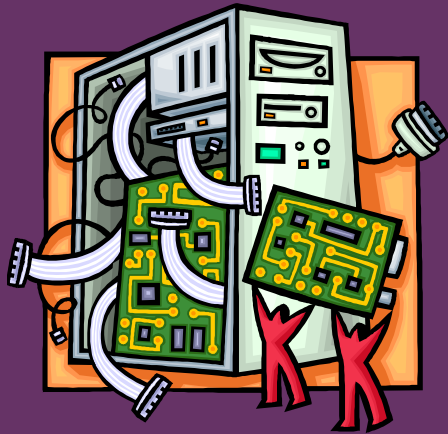


Figure 1.1 A Top-Down View of a Computer



There are four
main structural
components
of the computer:

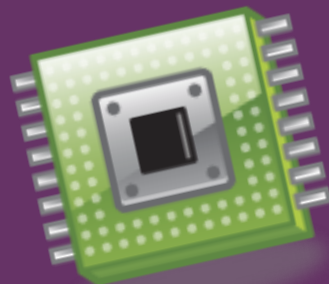
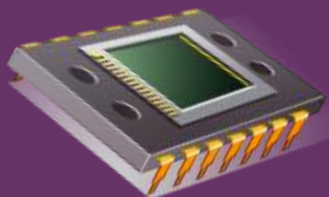


- ★ CPU – controls the operation of the computer and performs its data processing functions
- ★ Main Memory – stores data
- ★ I/O – moves data between the computer and its external environment
- ★ System Interconnection – some mechanism that provides for communication among CPU, main memory, and I/O



CPU

Major structural components:



- Control Unit
 - Controls the operation of the CPU and hence the computer
- Arithmetic and Logic Unit (ALU)
 - Performs the computer's data processing function
- Registers
 - Provide storage internal to the CPU
- CPU Interconnection
 - Some mechanism that provides for communication among the control unit, ALU, and registers



Multicore Computer Structure

- Central processing unit (CPU)
 - Portion of the computer that fetches and executes instructions
 - Consists of an ALU, a control unit, and registers
 - Referred to as a processor in a system with a single processing unit
- Core
 - An individual processing unit on a processor chip
 - May be equivalent in functionality to a CPU on a single-CPU system
 - Specialized processing units are also referred to as cores
- Processor
 - A physical piece of silicon containing one or more cores
 - Is the computer component that interprets and executes instructions
 - Referred to as a *multicore processor* if it contains multiple cores

Cache Memory

- Multiple layers of memory between the processor and main memory
- Is smaller and faster than main memory
- Used to speed up memory access by placing in the cache data from main memory that is likely to be used in the near future
- A greater performance improvement may be obtained by using multiple levels of cache, with level 1 (L1) closest to the core and additional levels (L2, L3, etc.) progressively farther from the core

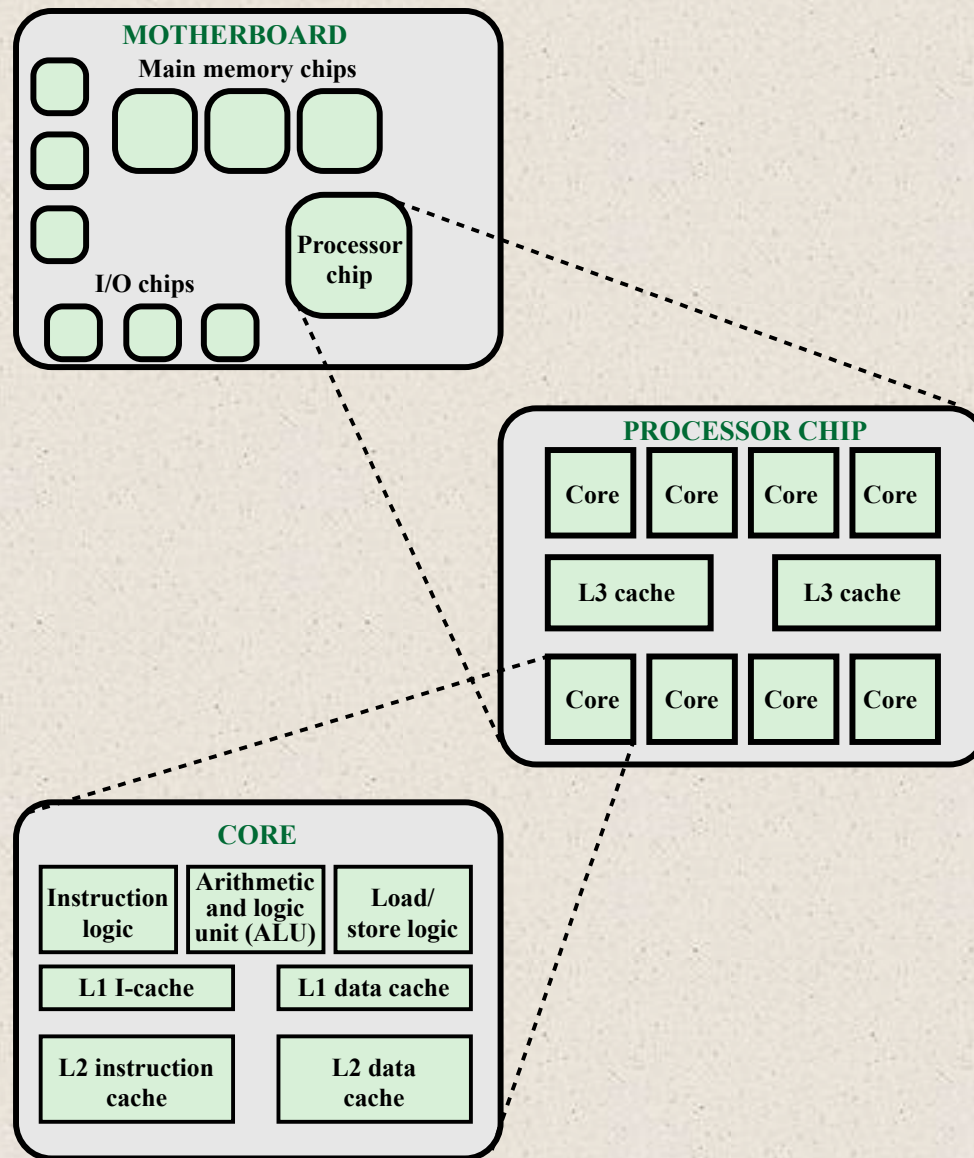
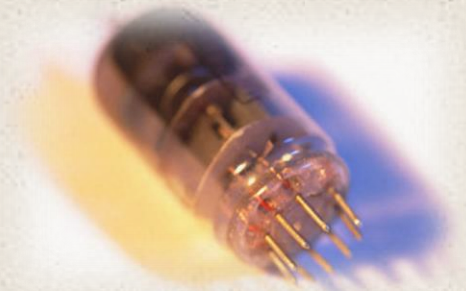


Figure 1.2 Simplified View of Major Elements of a Multicore Computer

+ History of Computers

First Generation: Vacuum Tubes

- Vacuum tubes were used for digital logic elements and memory
- IAS computer
 - Fundamental design approach was the stored program concept
 - Attributed to the mathematician John von Neumann
 - First publication of the idea was in 1945 for the EDVAC
 - Design began at the Princeton Institute for Advanced Studies
 - Completed in 1952
 - Prototype of all subsequent general-purpose computers



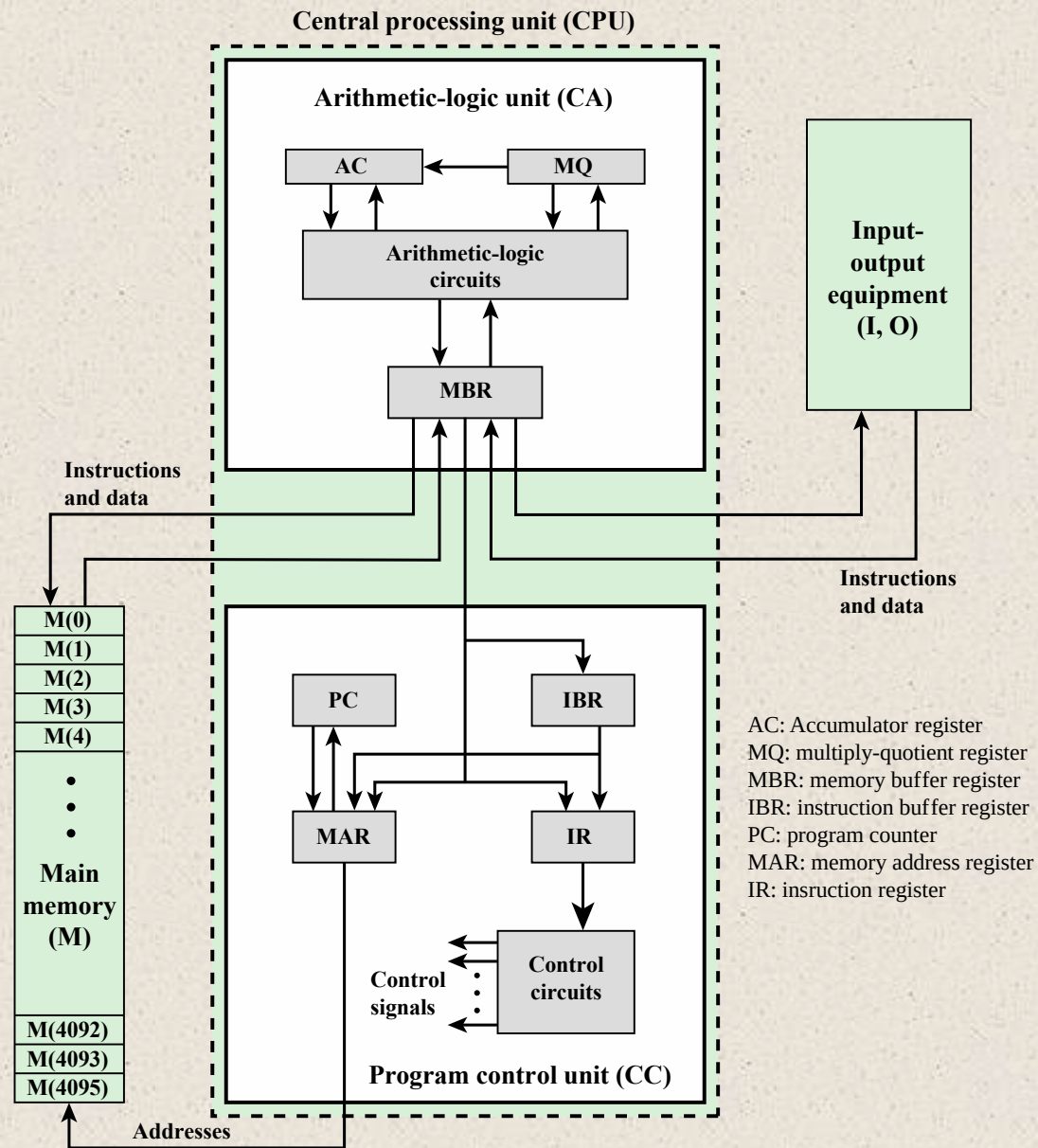


Figure 1.6 IAS Structure

Registers

Memory buffer register (MBR)

- Contains a word to be stored in memory or sent to the I/O unit
- Or is used to receive a word from memory or from the I/O unit

Memory address register (MAR)

- Specifies the address in memory of the word to be written from or read into the MBR

Instruction register (IR)

- Contains the 8-bit opcode instruction being executed

Instruction buffer register (IBR)

- Employed to temporarily hold the right-hand instruction from a word in memory

Program counter (PC)

- Contains the address of the next instruction pair to be fetched from memory

Accumulator (AC) and multiplier quotient (MQ)

- Employed to temporarily hold operands and results of ALU operations

+ History of Computers

Second Generation: Transistors

- Smaller
- Cheaper
- Dissipates less heat than a vacuum tube
- Is a *solid state device* made from silicon
- Was invented at Bell Labs in 1947
- It was not until the late 1950's that fully transistorized computers were commercially available

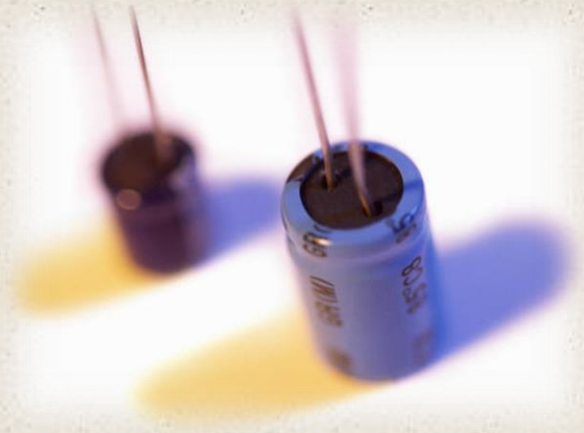


Table 1.2

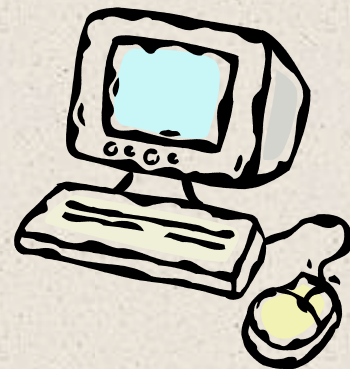
Computer Generations

Generation	Approximate Dates	Technology	Typical Speed (operations per second)
1	1946–1957	Vacuum tube	40,000
2	1957–1964	Transistor	200,000
3	1965–1971	Small and medium scale integration	1,000,000
4	1972–1977	Large scale integration	10,000,000
5	1978–1991	Very large scale integration	100,000,000
6	1991–	Ultra large scale integration	>1,000,000,000

Second Generation Computers

■ Introduced:

- More complex arithmetic and logic units and control units
- The use of high-level programming languages
- Provision of *system software* which provided the ability to:
 - Load programs
 - Move data to peripherals
 - Libraries perform common computations



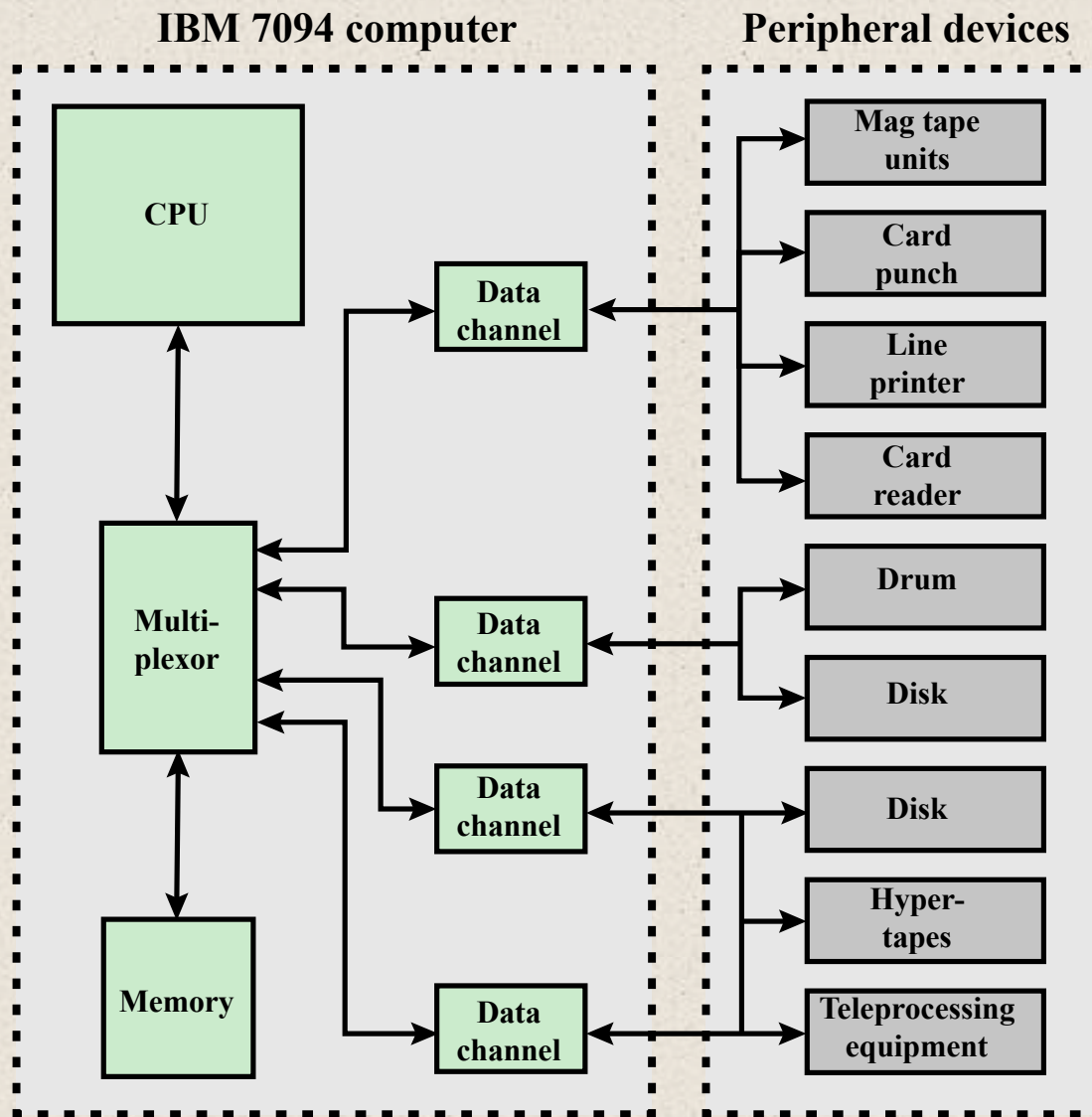


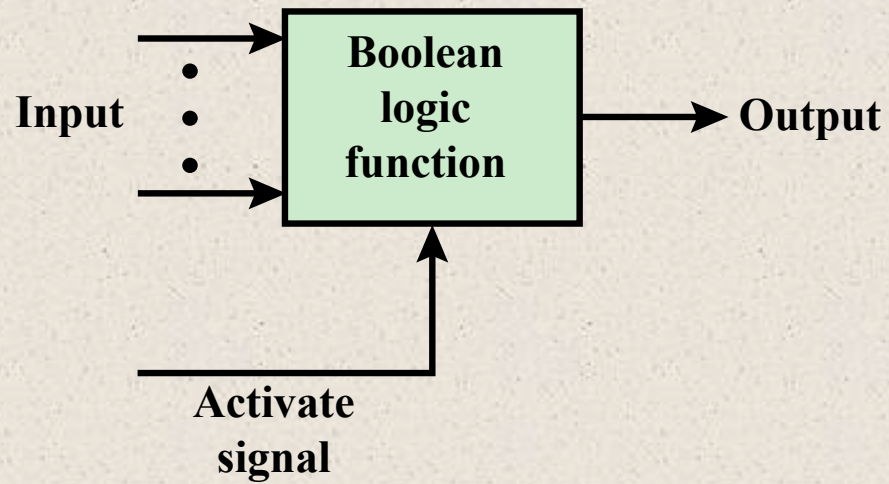
Figure 1.9 An IBM 7094 Configuration

History of Computers

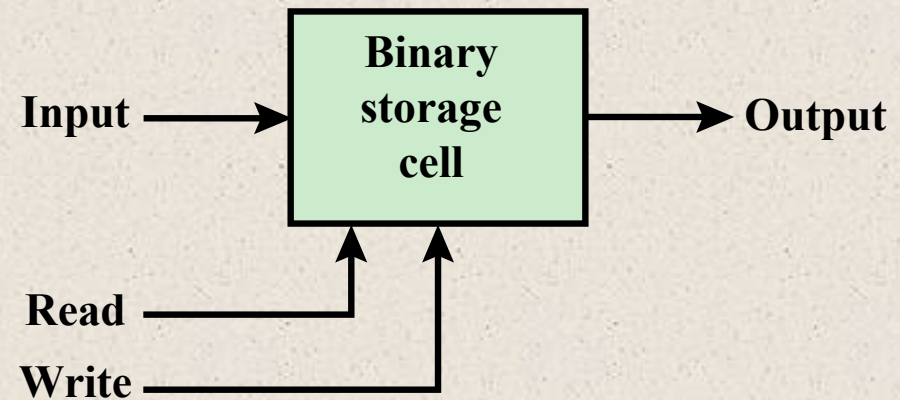
Third Generation: Integrated Circuits



- 1958 – the invention of the integrated circuit
- *Discrete component*
 - Single, self-contained transistor
 - Manufactured separately, packaged in their own containers, and soldered or wired together onto masonite-like circuit boards
 - Manufacturing process was expensive and cumbersome
- The two most important members of the third generation were the IBM System/360 and the DEC PDP-8



(a) Gate



(b) Memory cell

Figure 1.10 Fundamental Computer Elements

Integrated Circuits

- Data storage – provided by memory cells
 - Data processing – provided by gates
 - Data movement – the paths among components are used to move data from memory to memory and from memory through gates to memory
 - Control – the paths among components can carry control signals
- A computer consists of gates, memory cells, and interconnections among these elements
 - The gates and memory cells are constructed of simple digital electronic components
 - Exploits the fact that such components as transistors, resistors, and conductors can be fabricated from a semiconductor such as silicon
 - Many transistors can be produced at the same time on a single wafer of silicon
 - Transistors can be connected with a processor metallization to form circuits

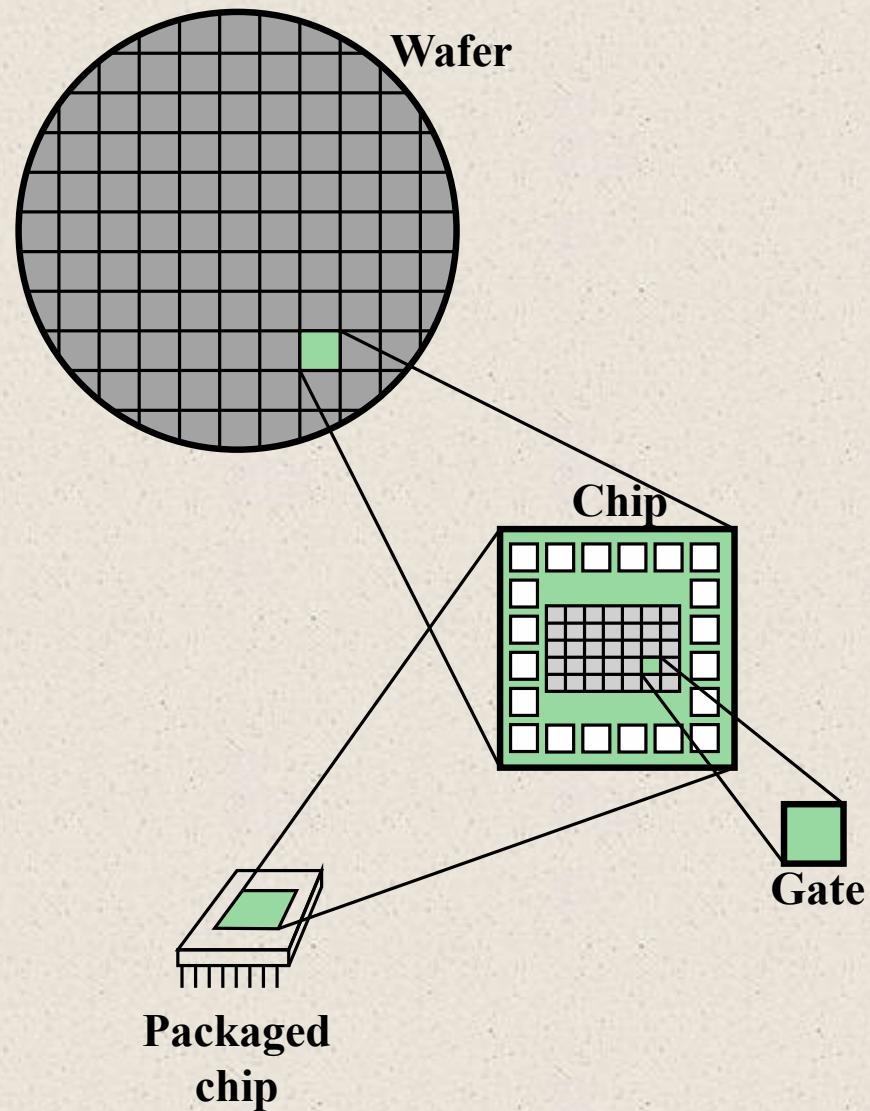


Figure 1.11 Relationship Among Wafer, Chip, and Gate

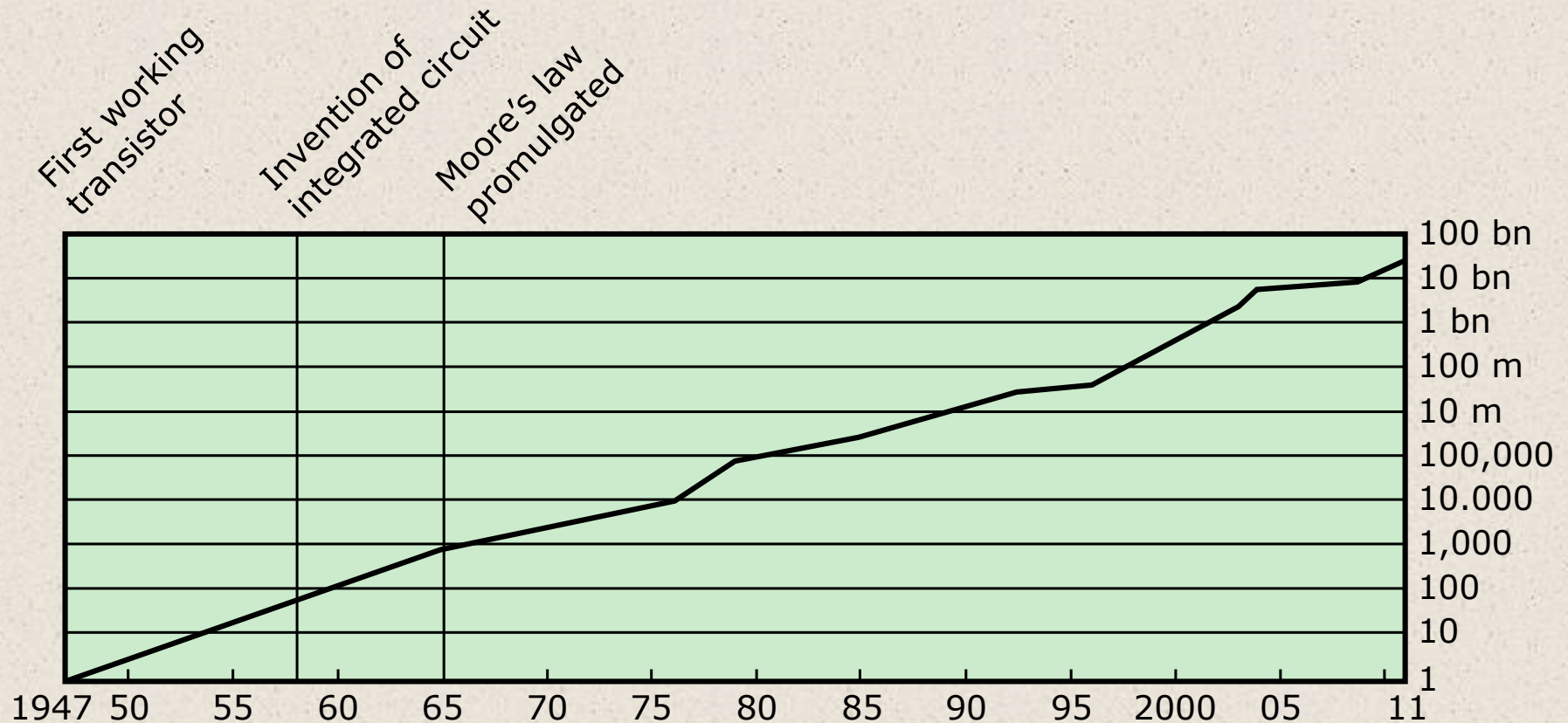


Figure 1.12 Growth in Transistor Count on Integrated Circuits (DRAM memory)

1965; Gordon Moore – co-founder of Intel

Observed number of transistors that could be put on a single chip was doubling every year

The pace slowed to a doubling every 18 months in the 1970's but has sustained that rate ever since

Consequences of Moore's law:

The cost of computer logic and memory circuitry has fallen at a dramatic rate

The electrical path length is shortened, increasing operating speed

Computer becomes smaller and is more convenient to use in a variety of environments

Reduction in power and cooling requirements

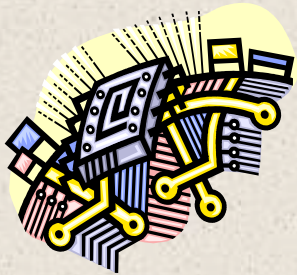
Fewer interchip connections



Later Generations

LSI
Large
Scale
Integration

VLSI
Very Large
Scale
Integration



Semiconductor Memory
Microprocessors

ULSI
Ultra Large
Scale
Integration

Evolution of Intel Microprocessors

	Pentium III	Pentium 4	Core 2 Duo	Core i7 EE 4960X
Introduced	1999	2000	2006	2013
Clock speeds	450 - 660 MHz	1.3 - 1.8 GHz	1.06 - 1.2 GHz	4 GHz
Bus width	64 bits	64 bits	64 bits	64 bits
Number of transistors	9.5 million	42 million	167 million	1.86 billion
Feature size (nm)	250	180	65	22
Addressable memory	64 GB	64 GB	64 GB	64 GB
Virtual memory	64 TB	64 TB	64 TB	64 TB
Cache	512 kB L2	256 kB L2	2 MB L2	1.5 MB L2/15 MB L3
Number of cores	1	1	2	6

(d) Recent Processors